Pinpoint: Product Requirements Document

Version 1.0 · Hackathon MVP · Team of three · Status: draft for review

1. Overview

Product: Pinpoint is a mobile-first web app that helps students learn faster by combining two proven techniques in one loop: speaking a topic aloud to expose gaps (active recall), and pinning each idea to a photo of a familiar place so it sticks (memory palace).

One-line pitch: *Explain it. Pin it. Own it.*

Platform: Progressive web app (React front end, Node and Express API, PostgreSQL database), with the Gemini API for content analysis.

2. Problem statement

Students spend hours re-reading notes, which feels productive but works poorly. They cannot tell which parts of a topic they truly understand, and what they do learn fades within days. Existing tools handle one piece in isolation (flashcards, notes, timers) and none connect checking understanding with long-term retention.

Problem	Effect on the student
Passive study (re-reading)	False sense of understanding
No feedback on what is missing	Gaps discovered only at the exam
Abstract facts are hard to retain	Forgetting within days
Building a memory palace mentally is hard	Most students never use the technique

3. Goals and non-goals

Goals

1. Let a student find out, in under three minutes, which key points of a topic they can and cannot explain.
2. Make a memory palace easy to build using photos of places the student already knows.

3. Test retention under a self-set time limit and show clearly what to revisit.
4. Give the student one dashboard that shows progress, tasks, and days left until the next exam.

Non-goals (MVP)

- Teacher or classroom features, live confusion tracking, or any multi-user collaboration.
- Social features, sharing of palaces, or leaderboards.
- Native mobile apps.
- Spaced-repetition scheduling, offline mode, and AI-graded voice answers in Recall.
- Support for uploads larger than two pages per topic.

4. Target users

Students worldwide, at any level, studying on a phone or laptop. One account type; no roles.

Persona	Need	How they use Pinpoint
Aarav, fast learner	Learn more in less time	Moves quickly through Explain, builds larger palaces, uses Recall to test speed
Meera, learns at her own pace	Understand deeply and remember it	Retries Explain until gaps close, pins only what she understands, repeats Recall

UI copy must say "learns at their own pace" and avoid the word "slow."

5. Core user flow

1. Sign up or log in, then land on the dashboard.
2. **Topic:** upload a PDF or photos (max 2 pages). Review the AI-extracted key points.
3. **Explain:** choose a topic and recording time, speak, and see the gap map and score.
4. Retry the gaps until the score reaches the pass mark.
5. **Palace:** add a place (photo), then add flags for each key idea.

6. **Recall:** play a round on a random place, with a time limit per flag, and review green and red flags.
7. **Home:** see tasks tick off, mastery rise, and the exam countdown.

Explaining precedes pinning by design: students should only memorize what they have shown they understand.

6. Functional requirements

Priority key: **P0** = required for the demo, **P1** = important, **P2** = nice to have.

6.1 Authentication and account

ID	Requirement	Priority
FR-1.1	A user can sign up with name, email, and password, and log in.	P0
FR-1.2	Passwords are stored hashed; sessions use a signed token.	P0
FR-1.3	All data is private to the account; no route returns another user's data.	P0
FR-1.4	A user can update their name and Explain pass mark (default 70%).	P2

6.2 Home (dashboard)

ID	Requirement	Priority
FR-2.1	Show the greeting "Hey {name}".	P0
FR-2.2	List previous units and topics with a mastery indicator; each topic shows whether its notes were uploaded or AI-generated.	P0
FR-2.3	Show an auto-generated task list per topic: Add material, Explain it, Pin it, Recall it, each marked complete or incomplete.	P0

ID	Requirement	Priority
FR-2.4	A task is complete when: a file exists (Add material); an explanation meets the pass mark (Explain it); at least one flag is linked to the topic (Pin it); the latest recall round is mostly green (Recall it).	P0
FR-2.5	Show a calendar with exam dates marked and one large number showing days left to the nearest exam.	P0 (number), P1 (full calendar)
FR-2.6	The countdown colour changes as the date nears (calm, amber, red).	P2
FR-2.7	A user can add, edit, and delete exams.	P0 (add), P2 (edit, delete)

6.3 Topic

ID	Requirement	Priority
FR-3.1	A user can upload PDFs or images for a topic.	P0
FR-3.2	The server enforces a maximum of two pages per topic (PDF page count, or two images) and returns a clear error otherwise.	P0
FR-3.3	On upload, the system extracts a title, a short notes summary, and 5 to 8 key points.	P0
FR-3.4	The user can edit the title, notes, and key points before and after saving.	P0
FR-3.5	A user can group topics into units.	P1
FR-3.6	The user can re-run extraction or remove a file.	P2

6.4 Explain

ID	Requirement	Priority
FR-4.1	The user selects a topic and a recording time (for example 1, 2, 3, or 5 minutes) with a visible countdown.	P0
FR-4.2	Speech is transcribed live in the browser; a typed-input fallback is always available.	P0
FR-4.3	The user can edit the transcript before submitting; submissions under roughly 15 words are blocked with a prompt to say more.	P1
FR-4.4	The system compares the transcript to the topic's key points and marks each as covered, partial, incorrect, or missing, with short feedback.	P0
FR-4.5	The score is computed in code: covered = 1, partial = 0.5, incorrect and missing = 0, shown as a percentage and as "You covered X of Y key points."	P0
FR-4.6	The user can retry only the unmet points.	P1
FR-4.7	A "Pin the gaps" action creates flags for unmet key points in the Palace.	P1
FR-4.8	Each attempt is saved so the user can see progress across attempts.	P1

6.5 Palace

ID	Requirement	Priority
FR-5.1	Show a grid of the user's places, each a photo tile with a name and flag count.	P0
FR-5.2	Opening a place shows its photo with multiple flags, with zoom and pan.	P0
FR-5.3	Tapping a flag opens a side panel (a bottom sheet on mobile) showing the flag's information, with an Edit button at the bottom.	P0

ID	Requirement	Priority
FR-5.4	"Add flag": the user taps a spot on the photo and enters the information; the flag may optionally link to a topic key point.	P0
FR-5.5	"Add place": the user takes or uploads a photo and names it.	P0
FR-5.6	Flag positions are stored as percentages of the image so they stay aligned on any screen size.	P0
FR-5.7	Photos are compressed on the device before upload.	P1
FR-5.8	A user can delete flags and places.	P1

6.6 Recall mode

ID	Requirement	Priority
FR-6.1	The app selects a random place from those the user has created.	P0
FR-6.2	The user sets the time allowed per flag before starting.	P0
FR-6.3	Flags are hidden at the start; the user taps one, tries to recall it, and reveals the answer.	P0
FR-6.4	A flag is **green** if recalled within the time limit and **red** if not (including timeout). Grading is self-confirmed by the student.	P0
FR-6.5	At the end of a round, the photo shows all flags coloured green or red.	P0
FR-6.6	Results update the topic's mastery and the Recall task on the dashboard.	P1
FR-6.7	Red flags are prioritized when choosing the next round.	P2

7. Non-functional requirements

Area	Requirement
Performance	Screens load in under 2 seconds on a typical mobile connection; an Explain result returns within about 6 seconds.
Reliability	If the AI service fails, a keyword-based fallback still returns a result; if speech recognition is unsupported, typed input is offered.
Security and privacy	Data is private by default; every query is scoped to the logged-in user; secrets (AI key, storage keys) live only on the server; room photos use non-guessable URLs.
Usability	Mobile-first layout, large tap targets, bottom-bar navigation with five tabs: Home, Topic, Explain, Palace, Recall.
Compatibility	Latest Chrome and Edge for voice; other modern browsers supported with typed input.
Accessibility	Sufficient colour contrast, and red and green states also use an icon or label so colour is never the only signal.
Cost control	AI calls are rate-limited per user and responses are cached during development.

8. AI behaviour requirements

Operation	Input	Output
Extract	Uploaded PDF or images	Title, notes summary, 5 to 8 key points with stable ids
Check explanation	Key points, notes, transcript	Per-point status, evidence, feedback, summary, and next-focus ids

Rules:

- Output must be JSON only and is validated by the server; invalid output is retried once.
- Judgement is based on meaning, not exact keywords, and only against the uploaded material.
- The prompt states that the transcript may contain speech-recognition errors.
- The score is never computed by the AI; the server calculates it.
- No personal details are included in prompts.

9. Data requirements

Main entities: users, units, topics, topic files, explanations, places, flags, recall rounds, recall results, and exams.

- Key points are stored on the topic, each with a stable id, and are referenced by flags and explanation results.
- Dashboard tasks are computed from existing records, not stored.
- Explanation attempts are kept so progress over time can be shown.

10. Success metrics

The hackathon goal is a working end-to-end demo. These are targets to measure on the team and a few test users, not guaranteed outcomes.

Metric	Target
Complete loop (upload, explain, pin, recall) works end to end in the demo	Yes, without errors
Time from upload to first gap map	Under 3 minutes
Self-measured recall improvement (quiz before vs after one explain-and-pin round)	A clear improvement on at least two testers, reported honestly with the method used
Explain result latency	Under 6 seconds
Test-user feedback: "I could see exactly what I was missing"	Majority agree

11. Dependencies and assumptions

Dependencies

- Gemini API key from Google AI Studio (a subscription to the Gemini app does not include API access).
- A hosted PostgreSQL database and an image and file storage service.
- Browser Web Speech API support for voice (Chrome and Edge).

Assumptions

- Students have a phone or laptop with a camera and microphone.
- Two pages per topic is enough for a first version.
- Free-tier limits are adequate for a hackathon demo; usage will be monitored.

12. Risks and mitigations

Risk	Mitigation
Voice recognition unsupported or inaccurate	Typed fallback; transcript editable before submission
AI returns invalid or wrong output	JSON validation and retry; judge only against uploaded material; keyword fallback
AI rate limits or outages during the demo	Per-user rate limits, caching, fallback, and a recorded backup demo
Slow photo uploads on mobile data	Client-side compression
Free-tier backend cold start	Warm-up ping before presenting
Privacy concerns around room photos and study notes	Private by default and non-guessable URLs; clearly stated in the pitch
Scope creep with three people working in parallel	Strict P0 list and cut order (section 13)

13. Release plan and scope control

Build order

Phase	Work
0	Project setup, authentication, schema, API contract, bottom-bar shell
1	Topic upload and key-point extraction (riskiest external call, so prove it first)
2	Explain: recording, AI check, score, and gap map
3	Palace: places grid, photo viewer, flags, side panel
4	Recall mode
5	Dashboard: greeting, tasks, countdown, calendar
6	Polish, seed demo data, deploy, record a backup demo

Cut order if time runs short

1. Full calendar grid (keep only the countdown number)
2. Manual tasks (keep only auto-generated ones)
3. Spaced repetition
4. Voice answers in Recall
5. Unit grouping (use a flat topic list)

The loop of upload, explain, pin, and recall must never be cut.

Suggested team split (three people)

Role	Owns	Hands off to
Person 1: Backend and AI	Schema, auth, uploads with the two-page limit, Gemini extraction and explain endpoints, scoring, recall and exam endpoints, deployment	Both front-end owners (API contract and seeded demo data)
Person 2: Front end, learning flow	App shell and bottom bar, Home dashboard, Topic screen, Explain screen with speech and typed fallback	Person 3 (topic and key-point data used for flags)
Person 3: Front end, Palace and Recall	Photo viewer with zoom and pan, flags and side panel, add place and add flag, Recall mode and timer, visual polish	Person 1 (flag and recall endpoints)

Agree the API routes and JSON shapes together in the first hour so all three can build in parallel against mock data. Share the demo script and backup recording: whoever finishes first records it, and all three rehearse it.

14. Future scope

- Spaced-repetition scheduling for flags
- Walk-through mode across a place's flags
- AI-checked spoken answers in Recall
- Recall speed tracking
- Read-only sharing of palaces for study groups
- Offline review
- Support for longer documents

15. Open questions

Question	Current default
Is Recall self-graded or AI-checked?	Self-graded with a timer
What is the Explain pass mark?	70%
Must every flag link to a key point?	No; allow both linked and free-note flags
Final product name	Pinpoint (alternatives: Loop, Stick, Recall)
Which storage provider?	Cloudinary or Supabase Storage

16. Demo script (about two minutes)

1. **Hook (15s):** "Re-reading doesn't work. Explaining does, and so does your own room."
2. **Explain (40s):** Explain a topic aloud, skipping one part on purpose; the gap map shows what is missing.
3. **Fix (15s):** Re-explain the missing part and watch the score rise.
4. **Pin (30s):** Photograph a desk and drop flags for the key ideas.
5. **Recall (15s):** Hide the flags, recall them against the timer, and one turns red.
6. **Close (5s):** "Explain it. Pin it. Own it."